

Notes: David & Venkateswaran (AER, 2019)

The Sources of Capital Misallocation

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1 What question does the paper answer?

The paper asks a very specific version of a broad misallocation question:

Observed cross-sectional dispersion in the average revenue product of capital (ARPK) is large. How much of that dispersion reflects (i) adjustment frictions, (ii) imperfect information/uncertainty, and (iii) other wedges or distortions?

The core contribution is a **moment-based decomposition** that separates different sources of ARPK dispersion using a **dynamic investment model** and a set of **investment moments** (serial correlation, volatility, and correlation with productivity shocks) together with **ARPK–productivity comovement**. Because different mechanisms move these moments in different ways, the paper provides a disciplined way to attribute measured misallocation to distinct channels.

The paper also makes a second contribution: it connects ARPK dispersion to **aggregate TFP losses** via an analytic approximation, so that the decomposition can be mapped into “how much each friction lowers aggregate productivity.”

2 Baseline environment and equilibrium objects

2.1 CES demand and revenue

There is a continuum of intermediate-good firms indexed by i . The final good is produced competitively with a CES aggregator with elasticity of substitution $\theta > 1$:

$$Y_t = \left(\int_0^1 \hat{Y}_{it}^{\frac{\theta-1}{\theta}} di \right)^{\frac{\theta}{\theta-1}}. \quad (1)$$

Final-good cost minimization implies the inverse demand curve

$$P_{it} = \left(\frac{\hat{Y}_{it}}{Y_t} \right)^{-\frac{1}{\theta}}. \quad (1)$$

Each firm has a constant-returns technology in *physical* output:

$$\hat{Y}_{it} = \hat{A}_{it} K_{it}^{\hat{\alpha}_1} N_{it}^{\hat{\alpha}_2}, \quad \hat{\alpha}_1 + \hat{\alpha}_2 = 1.$$

Combining technology with demand implies a *revenue function* of the form

$$P_{it} \hat{Y}_{it} = Y_t^{1/\theta} \hat{A}_{it} K_{it}^{\alpha_1} N_{it}^{\alpha_2}, \quad \alpha_j \equiv \left(1 - \frac{1}{\theta} \right) \hat{\alpha}_j, \quad (R)$$

so revenue has decreasing returns because the firm faces downward-sloping demand.

2.2 Capital adjustment costs

Capital evolves as installed capital after investment. Investment is costly via a convex adjustment cost:

$$\Phi(K_{it+1}, K_{it}) = \frac{\hat{\xi}}{2} \left(\frac{K_{it+1}}{K_{it}} - (1 - \delta) \right)^2 K_{it}, \quad (2)$$

where δ is depreciation and $\hat{\xi}$ is the primitive adjustment-cost parameter.

2.3 Firm problem and reduction to a one-state dynamic problem

Labor is chosen statically each period under full information and without wedges (baseline). Capital is chosen dynamically and is subject to wedges.

Let R_t be the real interest rate and T_{it+1}^K be a firm-specific wedge multiplying the effective purchase price of capital. The firm's dynamic problem is

$$V_{it} = \max_{K_{it+1}} \{ G A_{it} K_{it}^\alpha - \Phi(K_{it+1}, K_{it}) - R_t T_{it+1}^K K_{it+1} + \beta \mathbb{E}_{it}[V_{i,t+1}] \}, \quad (3)$$

where:

- A_{it} is a reduced-form productivity term that incorporates $\hat{A}_{it}$ and equilibrium aggregates (e.g. $Y_t^{1/\theta}$ and wages) after optimizing over labor,
- $\alpha \in (0, 1)$ is the curvature of operating profits in capital after labor is optimized out,
- $G > 0$ is a constant (from the static labor problem),
- $\beta \in (0, 1)$ is discounting.

2.4 Productivity and wedge processes

Let $a_{it} \equiv \log A_{it}$ and $k_{it} \equiv \log K_{it}$.

Productivity. Productivity follows an AR(1):

$$a_{i,t+1} = \rho a_{it} + \mu_{i,t+1}, \quad \mu_{i,t+1} \sim \mathcal{N}(0, \sigma_\mu^2). \quad (5)$$

Capital wedge. The (log) wedge in the cost of capital is decomposed into

$$\tau_{i,t+1}^K \equiv \log T_{i,t+1}^K = \varepsilon_{i,t+1} + \chi_i + \gamma a_{i,t+1}. \quad (6)$$

Components:

- $\varepsilon_{i,t+1}$: **transitory** wedge (i.i.d. over time),
- χ_i : **permanent** firm component (fixed effect),
- $\gamma a_{i,t+1}$: **correlated** wedge (systematically related to productivity).

The sign of γ matters: $\gamma < 0$ means more productive firms face *lower* effective capital costs.

2.5 Information friction and uncertainty

Firms do not observe the innovation $\mu_{i,t+1}$ when choosing $K_{i,t+1}$. Instead they observe a noisy signal

$$s_{i,t+1} = \mu_{i,t+1} + e_{i,t+1}, \quad e_{i,t+1} \sim \mathcal{N}(0, \sigma_e^2),$$

independent of μ . Then the posterior expectation is linear (Gaussian updating):

$$\mathbb{E}_{it}[\mu_{i,t+1} \mid s_{i,t+1}] = \left(\frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_e^2} \right) s_{i,t+1},$$

and the posterior variance (uncertainty) is

$$V \equiv \text{Var}_{it}(\mu_{i,t+1} \mid s_{i,t+1}) = \frac{\sigma_\mu^2 \sigma_e^2}{\sigma_\mu^2 + \sigma_e^2}. \quad (2)$$

Small V means high-quality information (low uncertainty); large V means noisy signals (high uncertainty).

3 Investment Euler equation, log-linearization, and policy rule

3.1 Log-linear Euler equation

The paper log-linearizes the Euler equation implied by (3). Let ξ denote a reparameterization of adjustment costs convenient for the log-linear approximation.¹

Under the information structure above, the log-linearized Euler equation can be written as

$$k_{i,t+1} \left((1 + \beta)\xi + 1 - \alpha \right) = (1 + \gamma)\mathbb{E}_{it}[a_{i,t+1}] + \varepsilon_{i,t+1} + \chi_i + \beta\xi \mathbb{E}_{it}[k_{i,t+2}] + \xi k_{it}. \quad (4)$$

Interpretation. Table 5 is an attribution exercise for China. Read it as:

- **Baseline (impose $\alpha + \gamma_k = 0.71$):** $\gamma_k = 0.00$, $\gamma = -0.70$. Correlated factors explain 47.4% of σ_{arpk}^2 (all attributed to productivity-dependent policies).
- **Allow moderate size dependence ($\alpha + \gamma_k = 0.54$):** $\gamma_k = -0.18$, $\gamma = -0.51$. Correlated factors explain 39.6%, split into size-dependent 14.2% and productivity-dependent 25.4%.
- **Allow strong size dependence ($\alpha + \gamma_k = 0.36$):** $\gamma_k = -0.36$, $\gamma = -0.33$. Correlated factors explain 39.8%, split into size-dependent 29.6% and productivity-dependent 10.2%.

What is robust: *some large correlated wedge remains no matter what*, but the decomposition between size vs. productivity dependence is not separately identified without extra structure.

(4) says that desired future capital increases with expected future productivity $\mathbb{E}_{it}[a_{i,t+1}]$, decreases with wedges (through $\varepsilon_{i,t+1}, \chi_i, \gamma a_{i,t+1}$), and is smoothed over time by adjustment costs (terms involving ξ and expectations of $k_{i,t+2}$).

¹The paper uses $\hat{\xi}$ in the primitive cost function (2), but uses ξ in the log-linear dynamics. They are monotone transforms; see the paper/appendix for mapping.

3.2 Closed-form log-linear capital policy

Solving the linear rational expectations system yields a law of motion:

$$\boxed{\begin{aligned} k_{i,t+1} = & \psi_1 k_{it} + \psi_2 \left((1 + \gamma) \rho a_{it} + \chi_i \right) + \psi_3 \varepsilon_{i,t+1} \\ & + \psi_4 \left((1 + \gamma) \rho a_{it} + \varepsilon_{i,t+1} + \chi_i + \mathbb{E}_{it}[\mu_{i,t+1}] \right). \end{aligned}} \quad (7)$$

The coefficients $(\psi_1, \psi_2, \psi_3, \psi_4)$ are pinned down by structural parameters:

$$\boxed{\begin{aligned} \xi(\beta\psi_1^2 + 1) &= \psi_1((1 + \beta)\xi + 1 - \alpha), \\ \psi_2 &= \frac{\psi_1}{\xi(1 - \beta\rho\psi_1)}, \quad \psi_3 = \frac{\psi_1}{\xi}, \quad \psi_4 = \frac{1 - \psi_1}{1 - \alpha}. \end{aligned}} \quad (8)$$

Why these coefficients matter.

- ψ_1 controls the persistence of capital (and thus the serial correlation of investment growth). Adjustment costs raise ψ_1 .
- ψ_4 is the loading on “news” about future productivity. Higher uncertainty (higher V) weakens the effective response of capital to that news.
- ψ_2 is a long-run sensitivity to persistent components like χ_i and expected productivity.

4 From ARPK dispersion to aggregate TFP

4.1 Average revenue product of capital (ARPK)

Define firm i 's average revenue product of capital:

$$\text{arpk}_{it} \equiv \log \left(\frac{P_{it} Y_{it}}{K_{it}} \right).$$

In the baseline, labor is frictionless and chosen optimally each period, so *dispersion in arpk summarizes capital misallocation*.

4.2 Aggregation result

Let aggregate output in logs be

$$y_t = a_t + \hat{\alpha}_1 k_t + \hat{\alpha}_2 n_t,$$

where k_t and n_t are aggregate log capital and labor, and a_t is aggregate TFP.

Under a log-normal approximation, aggregate TFP is

$$\boxed{a = a^* - \frac{(\theta\hat{\alpha}_1 + \hat{\alpha}_2)\hat{\alpha}_1}{2} \sigma_{\text{arpk}}^2, \quad \frac{da}{d\sigma_{\text{arpk}}^2} = -\frac{(\theta\hat{\alpha}_1 + \hat{\alpha}_2)\hat{\alpha}_1}{2}.} \quad (9)$$

Here a^* is the first-best TFP level (when ARPK is equalized across firms), and σ_{arpk}^2 is the cross-sectional variance of arpk_{it} .

4.3 Derivation sketch for (9)

The key steps mirror Hsieh–Klenow logic but exploit that (i) labor is frictionless and (ii) the CES structure yields log-linear aggregation under log-normality:

1. Under CES demand, revenues satisfy $P_{it}Y_{it} \propto \hat{A}_{it}K_{it}^{\alpha_1}N_{it}^{\alpha_2}Y_t^{1/\theta}$.
2. With frictionless labor, the firm-specific labor choice pins down N_{it} as a function of K_{it} and productivity, and thus arpk_{it} becomes a sufficient statistic for the firm-specific “effective distortion” in capital.
3. Aggregate output is a CES aggregator over firm quantities, which becomes a log-sum-exp aggregator over the relevant firm-specific object.
4. When that object (here, a linear function of arpk_{it}) is approximately normal across firms, the log-sum-exp equals its mean plus a constant times its variance; the variance term is exactly the misallocation penalty.

Thus, (9) gives a transparent mapping from *measured dispersion* to *aggregate loss*.

5 Identification in the random-walk case (Proposition 1)

The paper first builds intuition and identification in a tractable special case: $\rho = 1$ (productivity is a random walk) and ε_{it} is i.i.d.

In that case the cross-sectional moments of capital/investment and ARPK admit closed-form expressions in the parameters $(V, \xi, \gamma, \sigma_\varepsilon^2, \sigma_\chi^2)$.

5.1 Key empirical moments

Let k_{it} follow (7). Define:

- $\sigma_k^2 \equiv \text{Var}(k_{it})$,
- $\rho_{k,k-1} \equiv \text{Corr}(k_{it}, k_{i,t-1})$,
- $\rho_{k,a-1} \equiv \text{Corr}(k_{it}, a_{i,t-1})$,
- $\lambda_{\text{arpk},a}$: the regression coefficient of innovations in arpk on innovations in a (a measure of “correlated distortions” emphasized in the misallocation literature).

5.2 Closed-form expressions (random-walk case)

Under $\rho = 1$ the following hold:

$$\sigma_k^2 = \left(\frac{\psi_2^2}{1 - \psi_1^2} \right) (1 + \gamma)^2 \sigma_\mu^2 + \left(\frac{2\psi_3^2}{1 + \psi_1} \right) \sigma_\varepsilon^2. \quad (10)$$

$$\rho_{k,k-1} = \psi_1 - \frac{\psi_3^2 \sigma_\varepsilon^2}{\sigma_k^2}. \quad (11)$$

$$\rho_{k,a-1} = \left[\left(\frac{V}{\sigma_\mu^2} \right) (1 - \psi_1) + \psi_1 \right] \frac{\sigma_\mu \psi_2 (1 + \gamma)}{\sigma_k}. \quad (12)$$

$$\lambda_{\text{arpk},a} = 1 - (1 - \alpha)(1 + \gamma)\psi_2 \left(1 - \frac{V}{\sigma_\mu^2} \right). \quad (13)$$

5.3 Identification logic and proof sketch

Proposition 1 (One-to-one mapping). *In the random-walk case, the mapping from structural parameters $(V, \xi, \gamma, \sigma_\varepsilon^2, \sigma_\chi^2)$ to the reduced-form moments $(\sigma_k^2, \rho_{k,k-1}, \rho_{k,a-1}, \lambda_{\text{arpk},a})$ is one-to-one.*

Proof sketch / constructive identification. The key is that different parameters move the moments in different directions.

Step 1 (identify adjustment costs via persistence). The coefficient ψ_1 is pinned down by ξ through (8), and it enters (11) monotonically. Holding fixed the “noise” term $\psi_3^2 \sigma_\varepsilon^2 / \sigma_k^2$, higher ξ raises ψ_1 and thus raises $\rho_{k,k-1}$. Hence, variation in $\rho_{k,k-1}$ disciplines ξ .

Step 2 (separate transitory wedges from adjustment costs). Equation (10) shows that σ_k^2 loads on two additive pieces: a productivity-driven term (proportional to $(1 + \gamma)^2 \sigma_\mu^2$) and a transitory-wedge term (proportional to σ_ε^2). Given ψ_1 (thus ξ), the pair $(\sigma_k^2, \rho_{k,k-1})$ jointly identifies the contribution of σ_ε^2 , because (11) isolates $\psi_3^2 \sigma_\varepsilon^2 / \sigma_k^2$.

Step 3 (identify uncertainty via the responsiveness of investment to past productivity). Uncertainty enters only through the term V/σ_μ^2 in (12) and (13). Intuitively: when information is noisy (high V), firms respond less to innovations in productivity, weakening the link between k and lagged a .

Step 4 (identify correlated distortions via $\lambda_{\text{arpk},a}$). The coefficient $\lambda_{\text{arpk},a}$ in (13) rises with γ (in magnitude) because a more negative γ makes wedges systematically lower for high-productivity firms, changing how arpk comoves with a . Given ψ_2 (from ξ) and V (from Step 3), $\lambda_{\text{arpk},a}$ pins down γ .

Step 5 (identify permanent factors from the remaining variance). Finally, with $(\xi, V, \gamma, \sigma_\varepsilon^2)$ pinned down, the remaining component of σ_{arpk}^2 (in the quantitative model) is attributed to σ_χ^2 , the dispersion in permanent wedges.

A formal proof in the paper shows that the Jacobian of the mapping is nonsingular, and the “isomoment” curves intersect uniquely (visualized in Figure 1). $\square$

5.4 Figure 1: pairwise identification intuition

- **Panel A (ξ vs. γ):** both adjustment costs and correlated wedges move the dispersion of k , but only adjustment costs strongly affect persistence. Hence $(\sigma_k^2, \rho_{k,k-1})$ isolates (ξ, γ) .
- **Panel B (V vs. γ):** uncertainty and correlated wedges both affect the k - a relationship, but in different ways across $\rho_{k,a-1}$ and $\lambda_{\text{arpk},a}$, giving a unique intersection.
- **Panel C (σ_ε^2 vs. γ):** transitory wedges shift $\rho_{k,k-1}$ and $\lambda_{\text{arpk},a}$ differently than correlated wedges.
- **Panel D (V vs. ξ):** uncertainty affects how forward-looking firms are (response to productivity news), while adjustment costs affect smoothing/persistence.

6 Quantitative implementation

6.1 Data construction and empirical moments

For each country, the paper constructs:

- **Productivity:** a_{it} from firm value-added and capital (and implicitly labor, via the reduced-form profit function). It estimates (ρ, σ_μ^2) from the AR(1) regression (5).
- **Investment growth:**

$$\iota_{it} \equiv k_{i,t+1} - k_{it}.$$

Target moments include σ_ι^2 , $\rho_{\iota,\iota-1}$, and $\rho_{\iota,a-1}$.

- **Misallocation:** $\text{arpk}_{it} = \log(P_{it}Y_{it}/K_{it})$ and its variance σ_{arpk}^2 and correlation with a_{it} .

6.2 Estimation strategy

The paper chooses five parameters $(\xi, V, \gamma, \sigma_\varepsilon^2, \sigma_\chi^2)$ to match five moments:

$$(\rho_{\iota,a-1}, \rho_{\iota,\iota-1}, \sigma_\iota^2, \rho_{\text{arpk},a}, \sigma_{\text{arpk}}^2).$$

It uses simulated method of moments on the quantitative model.

6.3 Table 1: parameterization summary

Table 1 collects:

- externally set parameters $(\theta, \beta, \delta, \hat{\alpha}_1, \hat{\alpha}_2)$,
- productivity-process parameters (ρ, σ_μ^2) estimated from data,
- the five structural parameters to be estimated from the target moments.

6.4 Table 2: target moments

Key differences across countries:

- Investment growth is **more volatile** in China (σ_t^2 larger).
- Investment growth is **less persistent** (more negative autocorrelation) in China.
- ARPK dispersion is **much larger** in China.

6.5 Table and figure

Table 1 (Parameterization summary). This table is mainly a “map” of what is calibrated vs. estimated. What we keep in mind:

- Common calibrated values: $\theta = 6$, $\beta = 0.95$, $\delta = 0.10$.
- Input elasticities differ by country (used to translate ARPK dispersion into TFP losses via (9)):
 - United States: $(\hat{\alpha}_1, \hat{\alpha}_2) = (0.33, 0.67)$.
 - China: $(\hat{\alpha}_1, \hat{\alpha}_2) = (0.50, 0.50)$.
- Productivity process (ρ, σ_μ^2) is estimated directly from firm productivity data.
- The decomposition parameters $(\xi, V, \gamma, \sigma_\varepsilon^2, \sigma_\chi^2)$ are estimated by matching investment/ARPK moments.

Parameter	Description	Target/value
Preferences/production		
θ	Elasticity of substitution	6
β	Discount rate	0.95
δ	Depreciation	0.10
$\hat{\alpha}_1$	Capital share	0.33 US/0.50 China
$\hat{\alpha}_2$	Labor share	0.67 US/0.50 China
Productivity/frictions		
ρ	Persistence of productivity	$\rho_{a,a-1}$
σ_μ^2	Shocks to productivity	
V	Signal precision	$\rho_{a,a-1}$
ξ	Adjustment costs	
γ	Correlated factors	$\rho_{arpk,a}$
σ_ε^2	Transitory factors	
σ_χ^2	Permanent factors	σ_{arpk}^2

	ρ	σ_μ^2	$\rho_{a,a-1}$	$\rho_{\chi,a-1}$	$\rho_{arpk,a}$	σ_ε^2	σ_{arpk}^2
China	0.91	0.15	0.29	-0.36	0.76	0.14	0.92
United States	0.93	0.08	0.13	-0.30	0.55	0.06	0.45

Table 2 (Target moments). The quantitative estimation targets the following moments (we list the numbers because they help me sanity check the estimation):

- **China:** $\rho = 0.91$, $\sigma_\mu^2 = 0.15$, $\rho_{a,a-1} = 0.29$, $\rho_{\chi,a-1} = -0.36$, $\rho_{arpk,a} = 0.76$, $\sigma_\varepsilon^2 = 0.14$, $\sigma_{arpk}^2 = 0.92$.
- **United States:** $\rho = 0.93$, $\sigma_\mu^2 = 0.08$, $\rho_{a,a-1} = 0.13$, $\rho_{\chi,a-1} = -0.30$, $\rho_{arpk,a} = 0.55$, $\sigma_\varepsilon^2 = 0.06$, $\sigma_{arpk}^2 = 0.45$.

My read: China has *much larger* ARPK dispersion and investment volatility, and a stronger link between investment and lagged productivity.

Figure 1 (Pairwise identification, analytical). The figure plots pairs of isomoment curves in four parameter planes. The key intuition we take away: *different frictions “rotate” the isomoment curves differently*, so the intersection is unique. This is why the paper insists on using *multiple investment moments* rather than only σ_{arpk}^2 .

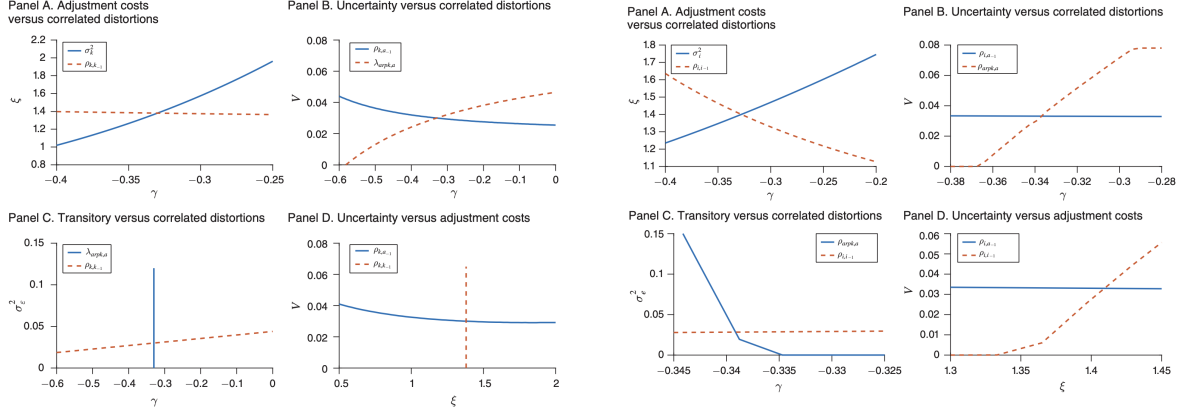


FIGURE 1. PAIRWISE IDENTIFICATION: ISOMOMENT CURVES

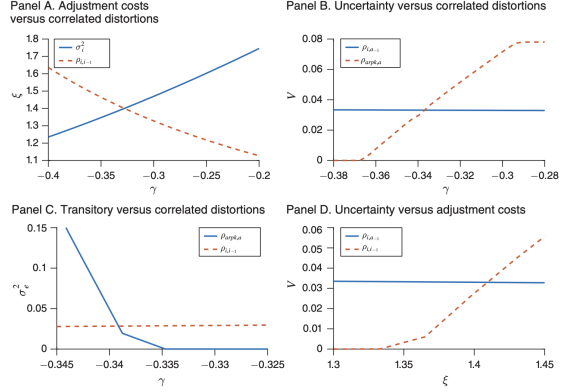


FIGURE 2. ISOMOMENT CURVES: QUANTITATIVE MODEL

Figure 2 (Pairwise identification, quantitative). Same visual logic as Figure 1, but computed in the quantitative model used for estimation. This reassures me that identification is not an artifact of the random-walk special case.

6.6 Baseline estimates and decomposition: Table 3

Table 3 reports estimated parameters and decomposes σ_{arpk}^2 and aggregate TFP losses into contributions from:

1. adjustment costs (ξ),
2. uncertainty (V),
3. other firm-specific factors, separated into correlated (γ), transitory (σ_{ε}^2), and permanent (σ_{χ}^2).

Main quantitative message (Table 3). Estimated parameters.

- **China:** $\xi = 0.13$, $V = 0.10$, $\gamma = -0.70$, $\sigma_{\varepsilon}^2 = 0.00$, $\sigma_{\chi}^2 = 0.41$.
- **United States:** $\xi = 1.38$, $V = 0.03$, $\gamma = -0.33$, $\sigma_{\varepsilon}^2 = 0.03$, $\sigma_{\chi}^2 = 0.29$.

Shares of ARPK dispersion explained (percent of σ_{arpk}^2).

- **China:** adjustment costs 1.3%, uncertainty 10.3%, correlated 47.4%, transitory 0.0%, permanent 44.4%.
- **United States:** adjustment costs 10.8%, uncertainty 7.3%, correlated 14.4%, transitory 6.3%, permanent 64.7%.

So (to myself): *in China, the big story is correlated + permanent wedges; in the US, the big story is permanent wedges.*

- **Adjustment costs explain little** of ARPK dispersion (about 1% in China, 11% in the US).
- **Uncertainty matters, but modestly** (about 10% in China, 7% in the US).
- **Other factors dominate.** In China, correlated and permanent components together explain about 90% of ARPK dispersion; in the US, permanent factors alone are the majority share.

Interpretation of the parameter estimates.

- China has *low* estimated adjustment costs (ξ small) but *strong* correlated wedges (γ very negative), suggesting that high-productivity firms face systematically *lower* capital costs.
- The US has *high* estimated adjustment costs (consistent with smoother investment dynamics) and smaller correlated wedges.

Parameters	Adjustment costs ξ	Uncertainty V	Other factors		
			Correlated γ	Transitory σ^2	Permanent σ^2
China	0.13	0.10	-0.70	0.00	0.41
United States	1.38	0.03	-0.33	0.03	0.29
$\Delta\sigma^2_{\text{arpk}}$					
China	0.01	0.10	0.44	0.00	0.41
United States	0.05	0.03	0.06	0.03	0.29
$\Delta\sigma^2_{\text{arpk}}/\sigma^2_{\text{arpk}}$					
China	1.3%	10.3%	47.4%	0.0%	44.4%
United States	10.8%	7.3%	14.4%	6.3%	64.7%
Δa					
China	0.01	0.08	0.38	0.00	0.36
United States	0.02	0.01	0.03	0.01	0.13

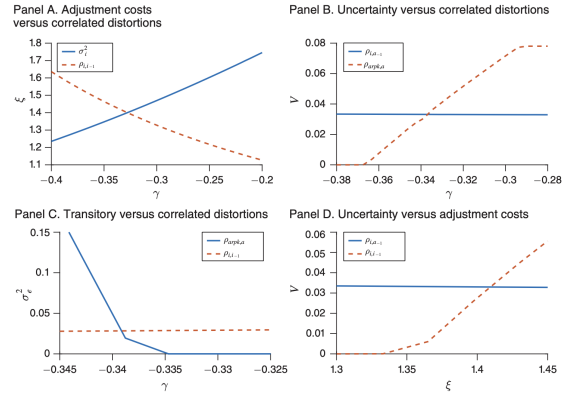


FIGURE 2. ISOMOMENT CURVES: QUANTITATIVE MODEL

6.7 Figure 2: isomoment curves in the quantitative model

Figure 2 replicates the identification intuition numerically in the fully specified model. Each panel plots two isomoment curves in a parameter plane; their intersection locates the parameter pair consistent with the data. The figure emphasizes that *even with richer quantitative features*, the same separation logic from Figure 1 holds.

7 Beyond the baseline: what are the “other factors”?

The baseline decomposition leaves a large residual attributed to γ and χ_i -type wedges. Section IV asks: can we map these “other factors” into more concrete economic mechanisms?

7.1 A. Heterogeneous markups and technologies (materials-based approach)

Introduce intermediate inputs M_{it} (materials). Let $\hat{\zeta}_{it}$ be the output elasticity of materials in gross output. Cost minimization with respect to M_{it} yields

$$\frac{P_{it}^M M_{it}}{P_{it} Y_{it}} = \frac{\hat{\zeta}_{it}}{\mu_{it}}, \quad (14)$$

where $\mu_{it} \equiv P_{it}/MC_{it}$ is the markup and P_{it}^M is the materials price.

Markup measurement. Given an estimate of $\hat{\zeta}_{it}$ (materials elasticity), (14) implies

$$\mu_{it} = \hat{\zeta}_{it} \cdot \frac{P_{it} Y_{it}}{P_{it}^M M_{it}}.$$

This is the De Loecker–Warzynski style logic: markups can be inferred from the ratio of revenue to expenditure on a flexible input.

7.2 B. Technology heterogeneity via ARPK–ARPN covariance

Allow for firm-heterogeneous capital elasticities $\hat{\alpha}_{it}$ and materials shares $\hat{\zeta}_{it}$. The paper shows:

$$\text{arpk}_{it} \equiv \log \left(\frac{P_{it} Y_{it}}{K_{it}} \right) = \log \left(\frac{P_{it}}{MC_{it}} \right) - \log \hat{\alpha}_{it} + \tau_{it}^K + \text{Constant}, \quad (15)$$

$$\text{arpn}_{it} \equiv \log \left(\frac{P_{it} Y_{it}}{N_{it}} \right) = \log \left(\frac{P_{it}}{MC_{it}} \right) - \log(\hat{\zeta}_{it} - \hat{\alpha}_{it}) + \tau_{it}^N + \text{Constant} \quad (16)$$

$$\approx \log \left(\frac{P_{it}}{MC_{it}} \right) + \left(\frac{\bar{\alpha}}{\hat{\zeta} - \bar{\alpha}} \right) \log \hat{\alpha}_{it} + \tau_{it}^N + \text{Constant}, \quad (17)$$

where $\tau_{it}^N = \log T_{it}^N$ is a labor wedge and $\bar{\alpha}$ is the average capital elasticity.

Key economic point. Markup variation shifts *both* ARPK and ARPN in the same direction (“scale” effect). Technology variation in $\hat{\alpha}_{it}$ shifts ARPK and ARPN in *opposite* directions (“mix” effect): higher $\hat{\alpha}_{it}$ lowers arpk but raises arpn.

7.3 Markup-adjusted average products and an upper bound on technology dispersion

Define markup-adjusted average products:

$$\widetilde{\text{arpk}}_{it} \equiv \text{arpk}_{it} - \log \left(\frac{P_{it}}{MC_{it}} \right), \quad \widetilde{\text{arpn}}_{it} \equiv \text{arpn}_{it} - \log \left(\frac{P_{it}}{MC_{it}} \right).$$

Proposition 2 (Upper bound on dispersion in $\log \hat{\alpha}_{it}$). *Suppose $\log \hat{\alpha}_{it}$ is uncorrelated with the*

wedges τ_{it}^K and τ_{it}^N . Then cross-sectional dispersion in $\log \hat{\alpha}_{it}$ satisfies

$$\sigma^2(\log \hat{\alpha}_{it}) \leq \frac{\sigma^2(\widetilde{\text{arpk}}) \sigma^2(\widetilde{\text{arpn}}) - \text{Cov}(\widetilde{\text{arpk}}, \widetilde{\text{arpn}})^2}{2 \left(\frac{\bar{\alpha}}{\bar{\xi} - \bar{\alpha}} \right) \text{Cov}(\widetilde{\text{arpk}}, \widetilde{\text{arpn}}) + \left(\frac{\bar{\alpha}}{\bar{\xi} - \bar{\alpha}} \right)^2 \sigma^2(\widetilde{\text{arpk}}) + \sigma^2(\widetilde{\text{arpn}})}. \quad (18)$$

Intuition of the bound. From (15)–(17), the pair $(\widetilde{\text{arpk}}, \widetilde{\text{arpn}})$ is driven by a “mix” component (technology: $\log \hat{\alpha}_{it}$) and a “scale” component (wedges: τ^K, τ^N). Technology induces *negative* comovement between the two, while common wedges can induce *positive* comovement. Given observed second moments of $(\widetilde{\text{arpk}}, \widetilde{\text{arpn}})$, the largest feasible role for technology arises when the wedge-driven part is maximally positively correlated (correlation = 1), which yields the bound (18). Thus, a more positive $\text{Cov}(\widetilde{\text{arpk}}, \widetilde{\text{arpn}})$ leaves less room for technology heterogeneity. $\square$

7.4 Table 4: interpreting the markup/technology accounting

Table 4 uses:

- a materials-based markup measure (via (14)),
- the covariance structure of $(\widetilde{\text{arpk}}, \widetilde{\text{arpn}})$,

to attribute part of σ_{arpk}^2 to (i) markup dispersion and (ii) an *upper bound* on technology dispersion.

Main message (Table 4). Contribution to σ_{arpk}^2 (levels and shares).

- **China:** markups contribute $\Delta \sigma_{\text{arpk}}^2 = 0.05$ (3.8%); technology (upper bound) contributes 0.30 (23.1%). Total markup+technology: 0.35 (26.9%).
- **United States:** markups contribute 0.06 (13.6%); technology (upper bound) contributes 0.18 (44.4%). Total markup+technology: 0.24 (58.1%).

So: technology dispersion could be a big part of “other factors” (especially in the US), but it cannot explain everything (especially in China).

- Markup dispersion explains a *small-to-moderate* share of σ_{arpk}^2 (about 4% in China and 14% in the US).
- Technology dispersion (upper bound) can be quantitatively important, especially in the US.

8 Size-dependent distortions and a liquidity-based microfoundation

8.1 A. Size-dependent wedge specification

Augment the wedge process with a size term:

$$\tau_{i,t+1}^K = \varepsilon_{i,t+1} + \chi_i + \gamma a_{i,t+1} + \gamma_k k_{i,t+1}.$$

Log-linearization yields

$$k_{i,t+1} \left((1 + \beta)\xi + 1 - \alpha - (1 + \gamma_k) \right) = (1 + \gamma) \mathbb{E}_{it}[a_{i,t+1}] + \varepsilon_{i,t+1} + \chi_i + \beta\xi \mathbb{E}_{it}[k_{i,t+2}] + \xi k_{it}. \quad (19)$$

Key implication. Only the *sum* $\alpha + \gamma_k$ is identified from the Euler equation. Hence, in the baseline, a negative γ could partly reflect a size-dependent wedge rather than a purely productivity-dependent wedge.

TABLE 4—HETEROGENEOUS MARKUPS AND TECHNOLOGIES

	China	United States
<i>Moments</i>		
$\sigma^2 \left(\log \frac{p_s Y_s}{p^s M_s} \right)$	0.05	0.06
$\text{cov}(\text{arpk}_s, \text{arpm}_s)$	0.41	0.23
$\sigma^2(\text{arpm}_s)$	1.37	0.52
$\sigma^2(\text{arpm}_s)$	0.76	0.35
<i>Estimated $\Delta\sigma_{\text{arpk}}$</i>		
Dispersion in markups	0.05 (3.8%)	0.06 (13.6%)
Dispersion in $\log \hat{\alpha}_s$	0.30 (23.1%)	0.18 (44.4%)
Total	0.35 (26.9%)	0.24 (58.1%)

Note: The values in parentheses in the bottom panel are the contributions to *arpk* dispersion expressed as a fraction of total σ_{arpk}^2 .

TABLE 5—SIZE VERSUS PRODUCTIVITY-DEPENDENT FACTORS

	Correlated factors			Adj. costs ξ
	Size-dependent γ_k	Prod.-dependent γ	Total	
$\alpha + \gamma_k = 0.71$ (<i>baseline</i>)				
Parameters	0.00	-0.70		0.13
$\Delta\sigma_{\text{arpk}}^2 / \sigma_{\text{arpk}}^2$	0.0%	47.4%	47.4%	1.3%
$\alpha + \gamma_k = 0.54$				
Parameters	-0.18	-0.51		0.21
$\Delta\sigma_{\text{arpk}}^2 / \sigma_{\text{arpk}}^2$	14.2%	25.4%	39.6%	2.3%
$\alpha + \gamma_k = 0.36$				
Parameters	-0.36	-0.33		0.29
$\Delta\sigma_{\text{arpk}}^2 / \sigma_{\text{arpk}}^2$	29.6%	10.2%	39.8%	3.2%

8.2 Table 5: re-attributing correlated factors to size-dependent policies (China)

Table 5 performs a counterfactual attribution exercise: for different imposed values of $\alpha + \gamma_k$, it re-estimates γ and ξ and recomputes contributions.

Interpretation. Table 5 is an attribution exercise for China. We read it as:

- **Baseline (impose $\alpha + \gamma_k = 0.71$):** $\gamma_k = 0.00$, $\gamma = -0.70$. Correlated factors explain 47.4% of σ_{arpk}^2 (all attributed to productivity-dependent policies).
- **Allow moderate size dependence ($\alpha + \gamma_k = 0.54$):** $\gamma_k = -0.18$, $\gamma = -0.51$. Correlated factors explain 39.6%, split into size-dependent 14.2% and productivity-dependent 25.4%.
- **Allow strong size dependence ($\alpha + \gamma_k = 0.36$):** $\gamma_k = -0.36$, $\gamma = -0.33$. Correlated factors explain 39.8%, split into size-dependent 29.6% and productivity-dependent 10.2%.

What is robust: *some large correlated wedge remains no matter what*, but the decomposition between size vs. productivity dependence is not separately identified without extra structure.

As one allows more size dependence (more negative γ_k), the estimated productivity dependence γ becomes less negative. However, the overall contribution of correlated factors remains large, implying that *some persistent correlated wedge is needed* regardless of whether it loads on size or productivity.

8.3 B. Liquidity / external finance costs as a microfoundation

The paper outlines a simple liquidity-based mechanism: firms face a marginal cost of liquidity/external finance that increases with scale. Let the marginal liquidity cost satisfy

$$\frac{\partial \Upsilon}{\partial K} = \nu(1 - \beta R)^\omega K^\omega, \quad (20)$$

where $\nu > 0$ scales the cost and $\omega > 0$ determines curvature.

This implies an approximate mapping to the size-dependent wedge parameter:

$$\gamma_k = \omega \nu (1 - \beta R)^\omega K^\omega. \quad (21)$$

Interpretation. Table 5 is an attribution exercise for China. We read it as:

- **Baseline (impose $\alpha + \gamma_k = 0.71$):** $\gamma_k = 0.00$, $\gamma = -0.70$. Correlated factors explain 47.4% of σ_{arpk}^2 (all attributed to productivity-dependent policies).
- **Allow moderate size dependence ($\alpha + \gamma_k = 0.54$):** $\gamma_k = -0.18$, $\gamma = -0.51$. Correlated factors explain 39.6%, split into size-dependent 14.2% and productivity-dependent 25.4%.
- **Allow strong size dependence ($\alpha + \gamma_k = 0.36$):** $\gamma_k = -0.36$, $\gamma = -0.33$. Correlated factors explain 39.8%, split into size-dependent 29.6% and productivity-dependent 10.2%.

What is robust: *some large correlated wedge remains no matter what*, but the decomposition between size vs. productivity dependence is not separately identified without extra structure.

Liquidity constraints can generate wedges that are *persistent* and *increasing in size*, making them quantitatively plausible candidates for the large residual “other factors” in the baseline decomposition.

9 Robustness and extensions

9.1 A. Nonconvex adjustment costs (Table 6)

The paper adds a fixed cost $\hat{\xi}_f$ of adjusting capital (an “inaction” region). This removes the ability to use perturbation methods; the model is solved by value function iteration and re-estimated by SMM, adding a sixth target moment: the inaction rate.

Conclusion from Table 6. With nonconvex adjustment costs, the estimated adjustment-cost parameters rise (because fixed costs create “lumpy” investment).

Shares of σ_{arpk}^2 explained (percent).

- **China:** convex adj. 6.5%, fixed adj. 0.8%, uncertainty 10.1%, correlated 35.6%, transitory 0.0%, permanent 47.7%.
- **United States:** convex adj. 13.0%, fixed adj. 1.1%, uncertainty 7.1%, correlated 11.5%, transitory 4.4%, permanent 64.4%.

So: allowing for lumpy investment increases the role of adjustment costs somewhat, but the residual persistent factors are still the main story.

Allowing for nonconvexities does *not* overturn the main message: adjustment costs (both convex and fixed) explain only a modest share of σ_{arpk}^2 , while other persistent factors remain dominant.

TABLE 6—NONCONVEX ADJUSTMENT COSTS

Parameters	$\hat{\xi}$	(ξ)	$\hat{\xi}_V$	V	γ	σ_ε^2	σ_χ^2
China	0.075	(0.51)	0.002	0.09	-0.64	0.00	0.44
United States	0.250	(1.70)	0.002	0.03	-0.30	0.02	0.29

 $\Delta\sigma_{\text{arpk}}^2/\sigma_{\text{arpk}}^2$

China (%)	6.5		0.8	10.1	35.6	0.0	47.7
United States (%)	13.0		1.1	7.1	11.5	4.4	64.4

Notes: The second column (in parentheses) reports the value of the normalized adjustment cost parameter, ξ , for purposes of comparison to Table 3. The mapping between ξ and $\hat{\xi}$ is given in expression (A.1) in the online Appendix.

TABLE 7—FRICTIONAL LABOR: CHINA

Moments	ρ	σ_μ^2	$\rho_{\chi,\varepsilon}$	$\rho_{\chi,\lambda}$	$\rho_{\text{arpk},\lambda}$	σ_ε^2	σ_{arpk}^2
Parameters	0.92	0.16	0.33	-0.36	0.81	0.14	0.94
			ξ	V	γ	σ_ε^2	σ_χ^2
			0.78	0.11	-0.68	0.04	0.30
Aggregate effects							
$\Delta\sigma_{\text{arpk}}^2$			0.12	0.11	0.48	0.04	0.30
$\Delta\sigma_{\text{arpk}}^2/\sigma_{\text{arpk}}^2$			12.8%	11.3%	51.2%	4.0%	32.2%
Δa			0.36	0.32	1.44	0.11	0.90

9.2 B. Frictional labor (Table 7)

The baseline assumes labor is frictionless and fully informed, making labor a static decision. The frictional-labor extension introduces wedges/frictions in labor that interact with capital and can amplify misallocation.

Conclusion from Table 7 (China). In the frictional labor extension, the estimated parameters for China are roughly: $\xi = 0.78$, $V = 0.11$, $\gamma = -0.68$, $\sigma_\varepsilon^2 = 0.04$, $\sigma_\chi^2 = 0.30$.

Shares of σ_{arpk}^2 explained (percent). Adjustment costs 12.8%, uncertainty 11.3%, correlated 51.2%, transitory 4.0%, permanent 32.2%.

The striking part is the implied *aggregate* effect: the table reports much larger aggregate TFP losses when labor is also frictional (misallocation now hits two margins).

Even with frictional labor, the contribution of correlated and permanent factors remains very large. Moreover, allowing labor frictions can increase implied aggregate TFP losses substantially, because misallocation now operates through both inputs.

9.3 C. Measurement error

Using Bils–Klenow–Ruane style logic, the paper estimates how much of measured σ_{arpk}^2 could be additive measurement error by running

$$\Delta va_{it} = \Phi \text{arpk}_{it} + \Psi \Delta k_{it} - \Psi(1 - \lambda) \text{arpk}_{it} \cdot \Delta k_{it} + D_{jt} + \epsilon_{it},$$

where D_{jt} are industry-year fixed effects. Estimates of λ close to 1 suggest limited measurement error contribution.

Finding. The paper finds $\lambda \approx 0.92$ in China and $\lambda \approx 0.88$ in the US, suggesting measurement error explains only about 10% of observed ARPK dispersion.

10 Figure 3: putting it all together

Figure 3 is the “executive summary” graphic: it takes the baseline decomposition (Table 3) and then overlays what can be explained by (i) markups and (ii) technology dispersion (upper bound, Table 4).

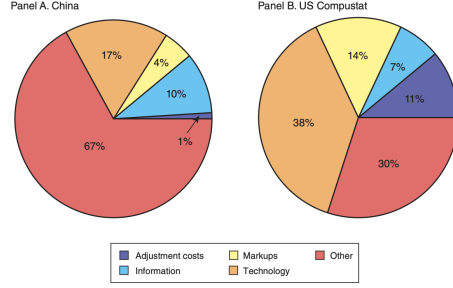


FIGURE 3. THE SOURCES OF MISALLOCATION

Note: The numbers for the contribution of technological dispersion denote the upper bound as calculated in footnote 43.

- **China:** the pie chart says roughly *Other* $\approx 67\%$, *Technology* $\approx 17\%$, *Information* $\approx 10\%$, *Markups* $\approx 4\%$, *Adjustment costs* $\approx 1\%$.
- **United States:** *Technology* $\approx 38\%$, *Other* $\approx 30\%$, *Adjustment costs* $\approx 11\%$, *Markups* $\approx 14\%$, *Information* $\approx 7\%$.

My interpretation: even after allowing for markups and technology, a large residual “Other” remains, especially in China, which keeps the door open to size-dependent policies, financial frictions, or other institutional wedges.

11 Short summary

1. **Dispersion in ARPK is not diagnostic by itself.** Different frictions can generate the same σ_{arpk}^2 .
2. **Dynamic moments identify mechanisms.** Adjustment costs, uncertainty, correlated wedges, and permanent wedges have distinct implications for investment volatility, persistence, and comovement with productivity.
3. **In the data, “other factors” dominate.** Adjustment costs and uncertainty explain only a minority share of ARPK dispersion.
4. **Persistent wedges are key.** Empirically plausible candidates include size-dependent policies, financial frictions, risk premia, or institutions generating firm fixed effects.
5. **Markup and technology heterogeneity explain some (not all) of the residual.** Markups are modest; technology can be larger (upper bound), especially in the US.

12 Conclusion

12.1 Summary

This paper’s central message is that **dispersion in (value-added) average revenue products of capital (ARPK, often used as a sufficient statistic for “capital misallocation”) is not a single**

object and cannot be mapped one-to-one into “distortions” without disentangling the underlying sources.

The authors develop a dynamic methodology that decomposes observed ARPK dispersion into distinct components:

1. **Adjustment frictions** (convex and nonconvex) that make capital respond sluggishly to shocks,
2. **Informational frictions / uncertainty** (firms invest under imperfect information about future productivity),
3. **Technology (and related heterogeneity)** that changes the mapping from inputs to output,
4. **Markup heterogeneity** (imperfect competition) that changes revenue products without changing physical marginal products,
5. **Other residual firm-specific factors**, captured as persistent wedges (including components correlated with productivity and fixed effects).

Empirically, the paper concludes that **the bulk of ARPK dispersion is not explained by standard dynamic investment frictions alone**. In Chinese manufacturing, adjustment costs and uncertainty account for only a modest share; most dispersion is driven by **persistent firm-specific factors** (“Other”) and a nontrivial technology component. In U.S. Compustat data, adjustment costs are more important than in China (especially for large firms and under alternative capital measurement), but technology and other factors remain major contributors, and markup heterogeneity is more quantitatively relevant than in China.

A broader implication is methodological: **many candidate frictions generate ARPK dispersion, but they leave different fingerprints in the data**, so one can use investment and capital dynamics to separate them and avoid attributing everything to “distortions.”

12.2 Key quantitative conclusions (headline numbers to remember)

Implied aggregate losses from dispersion. Using their estimated wedges, the paper reports substantial potential efficiency losses from misallocation-like dispersion:

$$\text{TFP losses} \approx 36\% \text{ in China, and } \approx 13\% \text{ in the United States.}$$

(Interpretation: these are losses relative to a benchmark without the distortionary components that generate dispersion in input products.)

Decomposition of ARPK dispersion (Figure 3). A compact way to remember the paper is the variance-share decomposition:

Source	China	U.S. Compustat
Adjustment costs	1%	11%
Information (uncertainty/news)	10%	7%
Technology (upper bound)	17%	38%
Markups	4%	14%
Other firm-specific factors	67%	30%

Interpretation:

- In **China**, the dominant driver is **Other** (persistent firm-specific wedges), not adjustment costs.
- In the **U.S.**, **technology-related heterogeneity** is much more important, and adjustment frictions matter more, but **Other** remains sizable.

12.3 Economic intuition: why these results fit together

The intuition follows from a simple accounting identity: observed ARPK dispersion can arise either because *(i)* firms are truly at different marginal products of capital due to wedges/distortions, or because *(ii)* revenue products differ for reasons that do not correspond to allocative inefficiency.

1. **Adjustment costs create dispersion even without distortions.** If firms face costs of changing capital, then after productivity shocks, some firms temporarily have “too much” or “too little” capital. This generates dispersion in ARPK through sluggish rebalancing.
2. **Uncertainty (informational frictions) creates dispersion because investment is state-contingent on beliefs.** When firms do not fully observe future productivity at the time of investing, optimal investment responds to signals/news, generating dispersion in realized ARPK that reflects information sets rather than policy wedges.
3. **Technology and markups can masquerade as “misallocation.”** If firms differ in production elasticities or in markups, then revenue-based products differ even if physical marginal products do not. A dispersion measure based on revenue products therefore mixes allocative wedges with technology/market-power heterogeneity.
4. **Persistent wedges are the “smoking gun” for true distortions in capital allocation.** A key empirical message is that the large residual component is *persistent* and partly *systematically related to productivity/size*. This pattern is difficult to generate from purely transitory shocks plus adjustment costs, and instead points to institutional/policy/financial forces that create long-lived differences in effective investment incentives across firms.

12.4 What the paper changes in how we interpret “misallocation”

- **Dispersion is not enough:** large ARPK dispersion does not automatically imply large distortionary wedges.

- **Dynamics discipline mechanisms:** different frictions imply different joint behavior of capital growth, investment rates, and input products, so the data can separate them.
- **Policy relevance:** distortions that are *persistent* and *systematically linked* to productivity/size are the most promising targets; purely transitory, size-orthogonal factors are less likely to drive large allocative losses.

12.5 Takeaway

Measured ARPK dispersion is a mixture of adjustment frictions, uncertainty, technology/markup heterogeneity, and persistent wedges; in China most dispersion comes from persistent “other” wedges, while in the U.S. technology and adjustment frictions matter more but residual wedges remain sizable, so decomposing the sources is essential before interpreting dispersion as policy-driven misallocation.